STATISTICS & MACHINE LEARNING · LESSON 8 OF 28

The scikit-learn Workflow: Train/Test Split, Baselines, Pipelines and Saving Models

Learn the standard machine-learning workflow in scikit-learn. The estimator API (fit, predict, transform), splitting data into train, validation and test sets with stratification and time order, baselines, ColumnTransformer and Pipeline to avoid leakage, comparing models, and saving and loading a trained pipeline.

ML class 4 · ✦ scikit-learn 1.7+ · Intermediate · ⏱ 110 min

By the end of this lesson you can

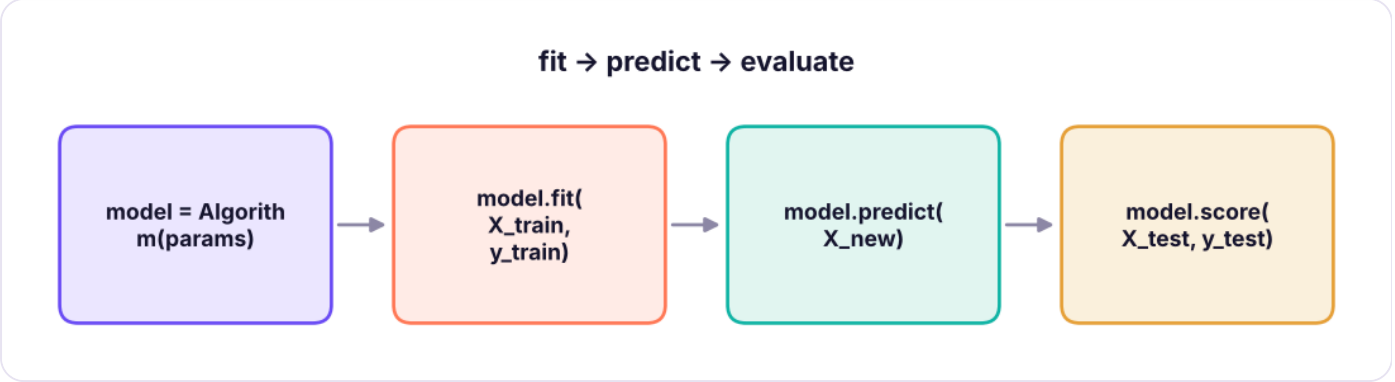
- Use the fit / predict / predict_proba / transform / score API consistently
- Split data into train and test (stratified, time-based) and explain why the test set is touched once
- Build dummy baselines and compare real models against them
- Combine preprocessing and a model in a Pipeline with ColumnTransformer
- Save, load and use a trained pipeline on new data

Datasets used

`telecom_churn.csv` `house_prices.csv`

The estimator API

Every scikit-learn model follows the same pattern, which makes swapping algorithms easy.



One consistent interface for every estimator.

Method	Available on	Does
`fit(X, y)`	Every estimator	Learns parameters from training data
`predict(X)`	Models	Predicted labels or values
`predict_proba(X)`	Most classifiers	Class probabilities
`transform(X)`	Transformers (scalers, encoders, PCA)	Converts features using fitted parameters
`fit_transform(X)`	Transformers	Fit then transform (training data only!)
`score(X, y)`	Models	Default metric (accuracy or R^2)
`get_params()` / `set_params()`	All	Hyperparameters

X is a 2-D table of features (DataFrame or array, shape *n_samples* × *n_features*); **y** is a 1-D target.

Train/test split

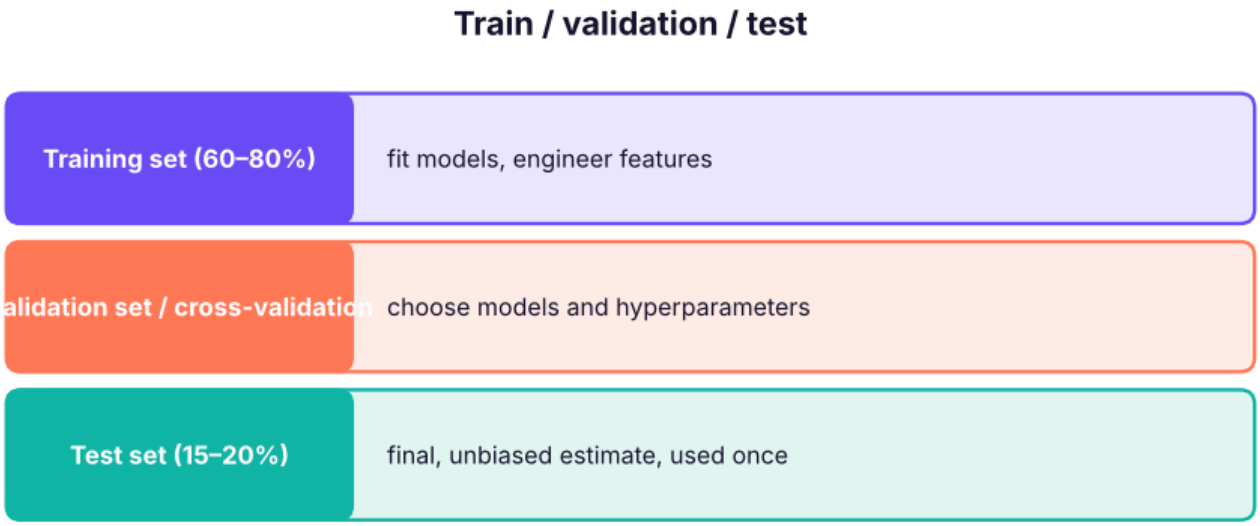
We judge a model on data it has **never seen**. Hold out a test set before doing anything else.

Python

```python
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split
tc = pd.read_csv("telecom_churn.csv")
X = tc.drop(columns=["customer_id", "churn", "total_charges"])          # total_charges ≈ tenure × monthly: redundant
y = (tc["churn"] == "Yes").astype(int)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
print(X_train.shape, X_test.shape, f"churn rate train {y_train.mean():.3f} | test {y_test.mean():.3f}")
```

OUTPUT

```
(2400, 7) (600, 7) churn rate train 0.258 | test 0.257
```



Typical data splits. The test set is a final exam: use it once, at the end.

Option	Use
`test_size=0.2`	Hold out 20%
`stratify=y`	Keep the class balance the same in both sets (classification)
`random_state=42`	Reproducible split
Time-based split	For time series and anything predicted in the future: train on the past, test on the most recent period
Group split (`GroupShuffleSplit`)	Keep all rows of a customer/store in one set

WATCH OUT **Never tune on the test set**

If you try many models and keep the one with the best **test** score, the test score is no longer an unbiased estimate. Choose with cross-validation on the training set (lesson 15); look at the test set once.

Baselines

A model is only useful if it beats a simple baseline.

Python

```
from sklearn.dummy import DummyClassifier, DummyRegressor
from sklearn.metrics import accuracy_score, recall_score
base = DummyClassifier(strategy="most_frequent").fit(X_train, y_train)
pred = base.predict(X_test)
print(f"always 'no churn': accuracy {accuracy_score(y_test, pred):.3f} | recall of churners
{recall_score(y_test, pred):.3f}")
rule = ((X_test.contract == "Month-to-month") & (X_test.tenure_months <= 12)).astype(int)
# business rule from the EDA
print(f"rule 'new month-to-month': accuracy {accuracy_score(y_test, rule):.3f} | recall
{recall_score(y_test, rule):.3f}")
```

OUTPUT

```
always 'no churn': accuracy 0.743 | recall of churners 0.000
rule 'new month-to-month': accuracy 0.777 | recall 0.240
```

74% accuracy by predicting "no churn" for everyone shows why **accuracy alone is misleading** for imbalanced problems (lesson 14).

Preprocessing with ColumnTransformer

Models need numbers on comparable scales. Numeric and categorical columns need different treatment.

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline, make_pipeline
num_cols = ["senior_citizen", "tenure_months", "monthly_charges", "support_calls"]
cat_cols = ["contract", "internet_service", "payment_method"]
prep = ColumnTransformer([
    ("num", make_pipeline(SimpleImputer(strategy="median"), StandardScaler()), num_cols),
    ("cat", make_pipeline(SimpleImputer(strategy="most_frequent"),
OneHotEncoder(handle_unknown="ignore")), cat_cols),
])
Xt = prep.fit_transform(X_train)
print(Xt.shape, list(prep.get_feature_names_out())[:7], "...")
```

OUTPUT

```
(2400, 14) ['num__senior_citizen', 'num__tenure_months', 'num__monthly_charges',
'num__support_calls', 'cat__contract_Month-to-month', 'cat__contract_One year',
'cat__contract_Two year'] ...
```

Pipelines: preprocessing + model as one object

Python

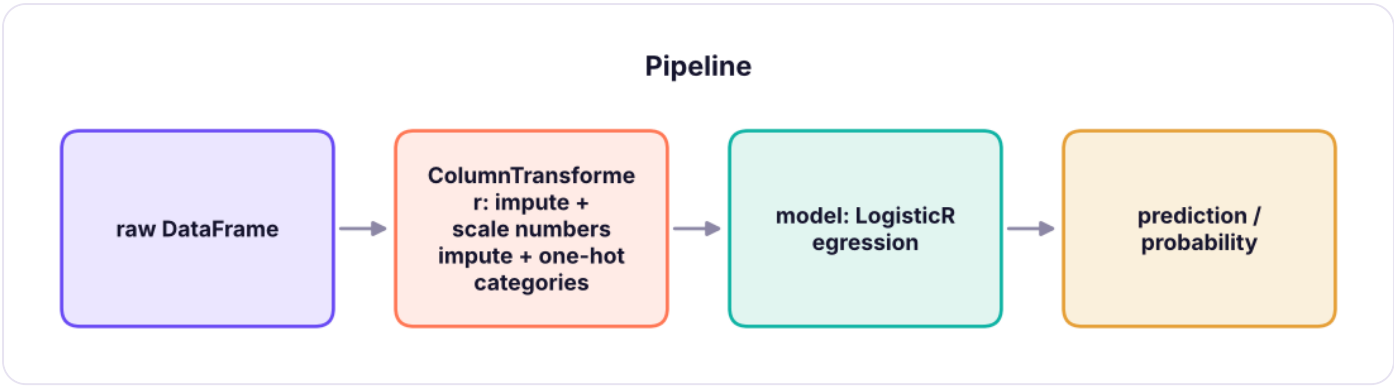
```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import roc_auc_score
models = {"logistic regression": LogisticRegression(max_iter=1000), "k-nearest neighbours":
KNeighborsClassifier(25),
          "random forest": RandomForestClassifier(300, min_samples_leaf=5, random_state=0),
"gradient boosting": HistGradientBoostingClassifier(random_state=0)}
for name, model in models.items():
    pipe = Pipeline([("prep", prep), ("model", model)]).fit(X_train, y_train)
    proba = pipe.predict_proba(X_test)[:, 1]
    print(f"{name:22s} accuracy {pipe.score(X_test, y_test):.3f} | ROC-AUC
{roc_auc_score(y_test, proba):.3f} | recall {recall_score(y_test,
pipe.predict(X_test)):.3f}")
```

OUTPUT

```
logistic regression    accuracy 0.790 | ROC-AUC 0.803 | recall 0.396
k-nearest neighbours   accuracy 0.785 | ROC-AUC 0.759 | recall 0.344
random forest          accuracy 0.798 | ROC-AUC 0.793 | recall 0.403
gradient boosting      accuracy 0.760 | ROC-AUC 0.755 | recall 0.455
```

Why pipelines?

1. **No leakage:** scalers and imputers learn their statistics from the training data only, inside each fit (and inside every cross-validation fold).
2. **One object** to tune, save and deploy: raw data in, predictions out.
3. **Less code** and fewer mistakes when switching models.



A pipeline chains steps; fit runs fit_transform on each step, predict runs transform then predict.

A regression example

Python

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
h = pd.read_csv("house_prices.csv")
Xh, yh = h.drop(columns=["house_id", "price_lakh"]), h["price_lakh"]
Xh_tr, Xh_te, yh_tr, yh_te = train_test_split(Xh, yh, test_size=0.2, random_state=1)
prep_h = ColumnTransformer([("num", make_pipeline(SimpleImputer(strategy="median"),
StandardScaler()), ["area_sqft", "bedrooms", "bathrooms", "age_years",
"distance_to_center_km", "parking"]),
                            ("cat", OneHotEncoder(handle_unknown="ignore"), ["city"])])
for name, m in [("mean baseline", DummyRegressor()), ("linear regression",
LinearRegression())]:
    p = Pipeline([("prep", prep_h), ("m", m)]).fit(Xh_tr, yh_tr).predict(Xh_te)
    print(f"{name:18s} MAE {mean_absolute_error(yh_te, p):6.2f} lakh | R² {r2_score(yh_te,
p):.3f}")
```

OUTPUT

```
mean baseline      MAE  23.53 lakh | R² -0.005
linear regression  MAE   7.18 lakh | R² 0.900
```

Saving and loading a model

Python

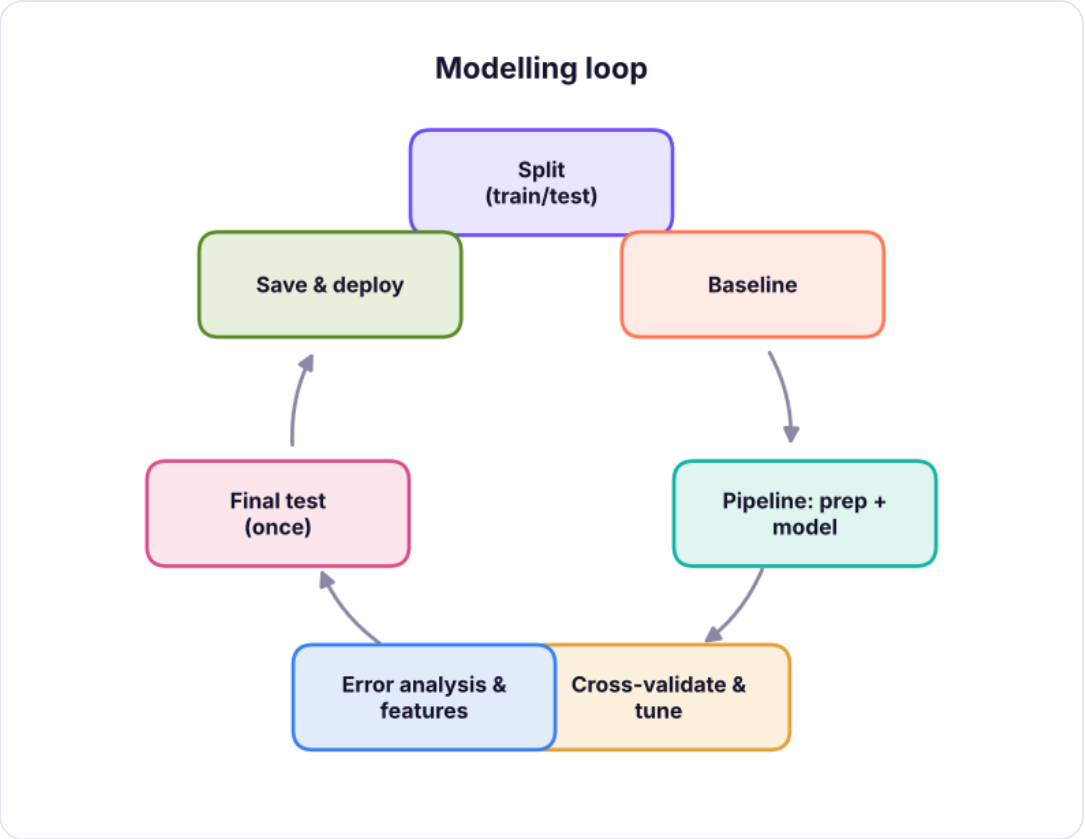
```
import joblib, os
best = Pipeline([("prep", prep), ("model",
LogisticRegression(max_iter=1000))]).fit(X_train, y_train)
joblib.dump(best, "churn_model.joblib")
loaded = joblib.load("churn_model.joblib")
new_customers = pd.DataFrame([{"senior_citizen": 0, "tenure_months": 3, "contract": "Month-
to-month", "internet_service": "Fiber", "payment_method": "UPI", "monthly_charges": 95.0,
"support_calls": 3},
                              {"senior_citizen": 1, "tenure_months": 60, "contract": "Two
year", "internet_service": "DSL", "payment_method": "Auto-debit", "monthly_charges": 55.0,
"support_calls": 0}])
print(loaded.predict_proba(new_customers)[:, 1].round(3), f"| file size
{os.path.getsize('churn_model.joblib') / 1024:.0f} KB")
os.remove("churn_model.joblib")
```

OUTPUT

```
[0.864 0.008] | file size 5 KB
```

Save the **whole pipeline**, note the library versions (`sklearn.__version__`), and keep a model card (data, date, metrics, intended use). Load models only from sources you trust: joblib/pickle files can run code.

The workflow, end to end



The modelling loop you will repeat in every lesson of this track.

Summary

KEY TAKEAWAYS

- Every estimator uses `fit`, then `predict` / `predict_proba` or `transform`; X is 2-D, y is 1-D.
- Split first (stratified for classification, by time for forecasting); the test set is used once.
- Always compare with a baseline (`DummyClassifier`, `DummyRegressor`, or a business rule).
- `ColumnTransformer` + `Pipeline` apply the right preprocessing to each column type and prevent leakage.
- Save the fitted pipeline with joblib and document versions and metrics.

Quick quiz

6 questions · tap an answer to check it

1. Why use `stratify=y` in `train_test_split` ?

- **A** To keep the same class proportions in train and test sets
- **B** To shuffle the data
- **C** To scale features
- **D** To remove outliers

Answer: A. Important for imbalanced classes.

2. Where should a StandardScaler learn its mean and standard deviation?

- **A** On the full dataset
- **B** On the training data only (inside the pipeline)
- **C** On the test data
- **D** It does not matter

Answer: B. Fitting on all data leaks test information.

3. A churn model has 74% accuracy and churn is 26%. What can you say?

- **A** It is good
- **B** It may be no better than predicting "no churn" for everyone; check recall and AUC
- **C** It is perfect
- **D** It is overfitting

Answer: B. Compare with a baseline.

4. What does `ColumnTransformer` do?

- **A** Applies different preprocessing to different columns and combines the results
- **B** Trains several models
- **C** Splits the data
- **D** Removes columns with missing values

Answer: A. e.g. scale numbers, one-hot encode categories.

5. Which split suits forecasting next month's demand?

- **A** Random split
- **B** Time-based: train on earlier periods, test on the latest period
- **C** Stratified split
- **D** No split

Answer: B. Random splits leak the future into training.

6. What should you save for deployment?

- **A** Only the model coefficients
- **B** The entire fitted pipeline (preprocessing + model) with version notes
- **C** The test set
- **D** The notebook only

Answer: B. One object turns raw data into predictions.

EXERCISE

Practice

(1) Build a pipeline for `hr_employees.csv` that predicts attrition with logistic regression; compare it with a dummy baseline using ROC-AUC. (2) Add `job_title` as a categorical feature: does AUC change? (3) Save and reload the pipeline and score two made-up employees.

Show solution (worked answer)

Python

```
hr = pd.read_csv("hr_employees.csv")
y2 = (hr.attrition == "Yes").astype(int); X2 = hr.drop(columns=["emp_id", "attrition",
"hire_date"])
nums = X2.select_dtypes("number").columns.tolist(); cats = [c for c in X2.columns if c
not in nums]
p2 = Pipeline([("prep", ColumnTransformer([("n", StandardScaler(), nums), ("c",
OneHotEncoder(handle_unknown="ignore"), cats)])), ("m",
LogisticRegression(max_iter=2000))])
a, b, c, d = train_test_split(X2, y2, test_size=.2, stratify=y2, random_state=0)
print(roc_auc_score(d, p2.fit(a, c).predict_proba(b)[:, 1]), roc_auc_score(d,
DummyClassifier().fit(a, c).predict_proba(b)[:, 1]))
```

INTERVIEW CORNER

Questions you may be asked

Q: Why do we need a train/test split? To estimate how the model performs on new, unseen data; training performance is optimistic because the model has seen those examples.

Q: What is a pipeline and why use one? A chain of preprocessing steps and a model trained and applied as one object; it prevents leakage, simplifies tuning and makes deployment reliable.

Q: What is the difference between fit_transform and transform? `fit_transform` learns parameters (e.g. mean, categories) and applies them, used on training data; `transform` applies already-learned parameters, used on validation, test and new data.

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